

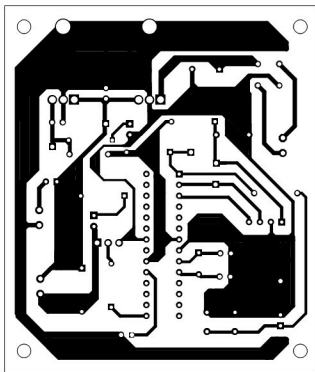


Fig. 6: An actual-size, single-side PCB layout for Android-powered access control via Bluetooth

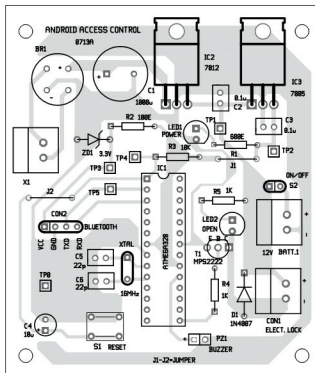


Fig. 7: Component layout for the PCB

causes it to open. The strike would remain locked in the event of power failure, but typically provision is made for the knob to open the door or provide alternative solution.

2. Fail-safe. In this configuration, applying electric current to the strike causes it to lock. It operates the same way as a magnetic lock would. If there

is a power failure, the door opens merely by being pushed or pulled.

Fail-secure configuration is used in this project. This is the most common type available in the market. It is very simple mechanically and electrically. It uses a solenoid to pull back a locking piece into the strike to allow opening of the door. If the solenoid doesn't pull back the locking piece, the strike won't open, creating a fail-safe situation. Normal operation of the doorknob or push bar from the system will still let people out in this situation.

There are various makes and models that use different voltages, both AC and DC. The electric strike used here operates off 12V DC. The whole system is equipped with a buzzer and an LED indicator to give the user audio and visual indication of door opening.

In this project, we have used a 12V battery as a back up source during power failure.

Software

The software is written in Arduino programming language and developed on Arduino 1.0.2 platform. When a valid password is received, the program makes pin 15 of the microcontroller high, which, in turn, unlocks the door. Here the password to unlock the door is '555.' You can change this password in the code as desired.

We have used a Freeduino board to burn the code into the microcontroller. You can also use an Arduino UNO board for the purpose.

To load the Arduino program

(also called Arduino sketch) into an ATmega328 MCU requires one of the following options:

1. Burn the sketch into the ATmega328 MCU that comes with the Arduino board
2. Purchase a fresh ATmega328 MCU and burn the sketch into it using Arduino board

The first option is straightforward. Burn the sketch (accesscontrol.ino) into the MCU, plug it out from Arduino board and use in your project. Note that, here, you need to sacrifice your Freeduino/Arduino board for this project.

The second option requires bootloader code in the MCU before loading the sketch into it. This option is useful if someone is working on a project using an ATmega328 MCU but does not want to sacrifice the Freeduino/Arduino board for use in another project. For this, you need a few additional components.

Two popular techniques available for burning the bootloader into fresh MCU using Arduino are: Optiloader and ArduinoISP.

Both techniques use Serial Peripheral Interface pins, require some steps and the complete steps to load the bootloader may take about five minutes. The Optiloader and Arduino ISP details can be found on the links:

<http://www.vwlowen.co.uk/arduino/bootloader/bootloader.htm> and
<http://oddwires.blogspot.in/2012/06/burning-bootloader-into-bare-atmega328.html>

After loading the bootloader, burn the Arduino sketch of this project into the MCU. There is also a third option. Purchase an ATmega328 MCU with bootloader, burn the sketch into it and then use it in your project. There are so many vendors available in India.

Construction and testing

An actual-size, single-side PCB layout for Android-powered access control via Bluetooth is shown in Fig. 6 and its component layout in Fig. 7. We have used a 28-pin, DIP package ATmega328 microcontroller in this